Ecosystem - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

Anmol Sharma • Lesson 30 • Feb 8, 2025

Ecosystem

Title: Ecosystem

1. Match the columns I and II

Column I	Column II
A Terrestrial ecosystem	P Aquarium
B Aquatic ecosystem	Q Grassland
C Man made ecosystems	R Estuary

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

2. Select the correct answer

I. Interaction of biotic and abiotic components result in a physical structure that is characteristic for each type of ecosystem.

II. Identification and enumeration of plant and animal species of an ecosystem gives its stratification.

(1) Only I is correct
(2) Only II is correct
(3) Both are correct
(4) None are correct

Title: Ecosystem

3. Statement A : In pond ecosystem, the abiotic component is the water with all the dissolved inorganic and organic substances and the rich soil deposit at the bottom of the pond.

Statement B : The solar input, the cycle of temperature, day-length and other climatic conditions regulate the rate of function of the entire pond.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

4. Match the columns I and II

Column I	Column II
A Consumers	P Mineral cycling
B Decomposers	Q Own food
C Autotrophs	R Depend on others

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

Title: Ecosystem

5. The stratification pattern in a community is well illustrated by a

(1) Aquatic community

(2) Desert community

(3) Forest community

(4) Grassland community

Title: Ecosystem

6. Select the correct answer

i. Conversion of inorganic into organic material with the help of the radiant energy of the sun by the autotroph

ii. Consumption of the autotrophs by heterotrophs;

iii. Decomposition and mineralisation of the dead matter to release them back for reuse by the autotrophs

The above events occur in a

(1) Population (2) Individual

(3) Ecosystem (4) Only pond

Title: Ecosystem

7. Which of the following statement is true?

(1) GPP utilizes of 1 – 5% of PAR

(2) GPP utilizes of 2-10% incident radiation

(3) NPP utilizes of 1.6-8% incident radiation

(4) GPP utilizes of 2-10% PAR

Title: Ecosystem

8. How much of the net primary productivity of a terrestrial ecosystem is eaten and digested by herbivores?

(1) 1%

(2) 10%

(3) 40%

(4) 90%

9. The rate of production of organic matter by consumers is

(1) Secondary productivity

(2) Net primary productivity

(3) Primary productivity

(4) Gross primary productivity

10. Secondary productivity is rate of formation of new organic matter by

(1) Consumers

(2) Decomposers

(3) Producers

(4) Parasites

Title: Ecosystem

11. The annual net primary productivity of the biosphere is approximately

(1) 170 billion tons

(2) 55 billion tons

(3) 170 million tons

(4) 55 million tons

Title: Ecosystem

12. Read the Statement-A and Statement-B carefully to mark the correct options given below:

Statement A : The annual net primary productivity of the whole biosphere is approximately 170 billion kilogram (dry weight) of organic matter.

Statement B : Despite occupying about 70 per cent of the surface, the productivity of the oceans are only 55 million tons.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

13. Statement A : Net primary productivity is greater than the gross primary productivity.

Statement B : A part of net primary productivity goes waste during respiration.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

14. In the equation GPP – R = NPP, R represents:

(1) Radiant energy

(2) Retardation factor

(3) Environment factor

(4) Respiration losses

Title: Ecosystem

15. Productivity is the rate of production of biomass expressed in terms of:

i. $(kcal\ m^{-3})yr^{-1}$

ii. g−2 yr−1

iii. g−1 yr−1

iv. $(kcal\ m^{-2})yr^{-1}$

(1) ii (2) iii

(3) ii and iv (4) i and iii

Title: Ecosystem

16. Statement A : Net primary productivity plus respiration losses (R), is the gross primary productivity (GPP).

Statement B : Net primary productivity is the available biomass for the consumption to autotrophs.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

17. Read the given statements and select the correct options

I. The amount of biomass or organic matter produced per unit area over a time period in plants during photosynthesis is called primary production.

II. Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.

III. Productivity is expressed in $gm^{-2}yr^{-1}$ or $(kcal\ m^{-2})yr^{-1}$

IV. Net primary productivity is the available biomass for the consumption to heterotrophs

(1) I, II and III (2) II and IV

(3) I and IV (4) I, II, III and IV

Title: Ecosystem

18. The annual net primary productivity of the whole biosphere is approximately____ tons (dry weight) of organic matter. Of this, despite occupying about____ of the surface, the productivity of the oceans are only____ tons.

(1) 55 billion, 70 per cent, 55 billion

(2) 170 million, 70 per cent, 55 million

(3) 55 million, 70 per cent, 55 million

(4) 170 billion, 70 per cent, 55 billion

Title: Ecosystem

19. The following table shows net productivities of various trophic levels in one ecosystem in Silver Springs, Florida receiving sunlight at a rate of 1, 700, 000 kcal/m^2/yr.

Primaryproducers 7168 kcal/m^2/yr

Primaryconsumers 1103 kcal/m^2/yr

Secondaryconsumers111 kcal/m^2/yr Tertiaryconsumers 5 kcal/m^2/yr

The efficiency energy transfer is the highest between

(1) Primary producers and primary consumers

(2) Primary consumers and secondary consumers

(3) Solar energy and primary producers

(4) Secondary consumers and tertiary consumers

20. Net primary productivity is equal to

(1) Organic matter synthesised by photosynthesis only

(2) Organic matter synthesised by photosynthesis minus utilisation in respiration and other losses

(3) Rate of respiration

(4) Rate of respiration + photosynthesis

21. The process of breaking down complex organic matter into its basic inorganic substances is called

(1) Humification

(2) Mineralisation

(3) Decomposition

(4) Leaching

22. Which of the following processes can slow down the process of decomposition?

(1) Anaerobiosis

(2) Aerobiosis

(3) Photosynthesis

(4) Respiration

23. Water soluble inorganic nutrients go down into soil horizon and get precipitated as unavailable salts through

(1) Humification

(2) Leaching

(3) Stratification

(4) Catabolism

Title: Ecosystem

24. Word detritus includes

(1) Dead plant parts

(2) Dead animal parts

(3) Animal waste

(4) All of these

Title: Ecosystem

25. Among the following, where do you think the process of decomposition would be the fastest?

(1) Tropical rain forest

(2) Antarctic

(3) Dry arid region

(4) Alpine region

Title: Ecosystem

26. Statement A : The process by which water soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts is called as leaching.

Statement B : The enzymatic process by which degraded detritus is converted into simpler inorganic substances is called catabolism.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

27. Which of the following factor doesnot influence the activity of detritivores and decomposer microbes

(1) Temperature

(2) Soil moisture

(3) Amount of organic matter in detritus

(4) Aerobic condition

28. Breakdown of detritus into smaller particles by detritivores is a process called

(1) Leaching

(2) Fragmentation

(3) Mineralisation

(4) Catabolism

Title: Ecosystem

29. Select the correct answer

A. Warm and moist environment favour decomposition

B. Low temperature and anaerobiosis inhibit decomposition resulting in clearance of organic materials.

(1) Only A is correct

(2) Only B is correct

(3) Both are correct

(4) None are correct

Title: Ecosystem

30. Which of the following is called detrivore?

(1) An animal feeding on a plant

(2) An animal feeding on another animal

(3) An animal feeding on decaying organic matter

(4) Plant feeding on an animal

Title: Ecosystem

31. Statement A : Decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and watersoluble substances like sugars.

Statement B : Low temperature and anaerobiosis inhibit decomposition resulting in build up of organic materials.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

32. Given below are two statements :

Statement I : Decomposition is a process in which the detritus is degraded into simpler substances by microbes.

Statement II : Decomposition is faster if the detritus is rich in lignin and chitin.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

33. Assertion : Humus undergoes decomposition at an extremely fast rate.

Reason : Various microbes can degrade humus effectively.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

Title: Ecosystem

34. Arrange sequentially the step of decomposition of detritus in an ecosystem

(A) Fragmentation

(B) Catabolism

(C) Mineralisation

(D) Leaching

(E) Humification

(1) A, D, B, E, C

(2) A, B, D, E, C

(3) A, D, E, B, C

(4) A, B, E, D, C

Title: Ecosystem

35. Statement A : The rate of decomposition is controlled by chemical composition of detritus and climatic factors.

Statement B : Decomposition is an anaerobic process.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

36. The organisms which physically and chemically break the complex dead organic remains are known as

(1) Saprotrophs

(2) Decomposers

(3) Both 1 and 2

(4) Parasites

Title: Ecosystem

37. Decomposers like fungi and bacteria are :-

i. Autotrophs

ii. Heterotrophs

iii. Saprotrophs

iv. Chemo-autotrophs

(1) i and iii

(2) i and iv

(3) ii and iii

(4) i and ii

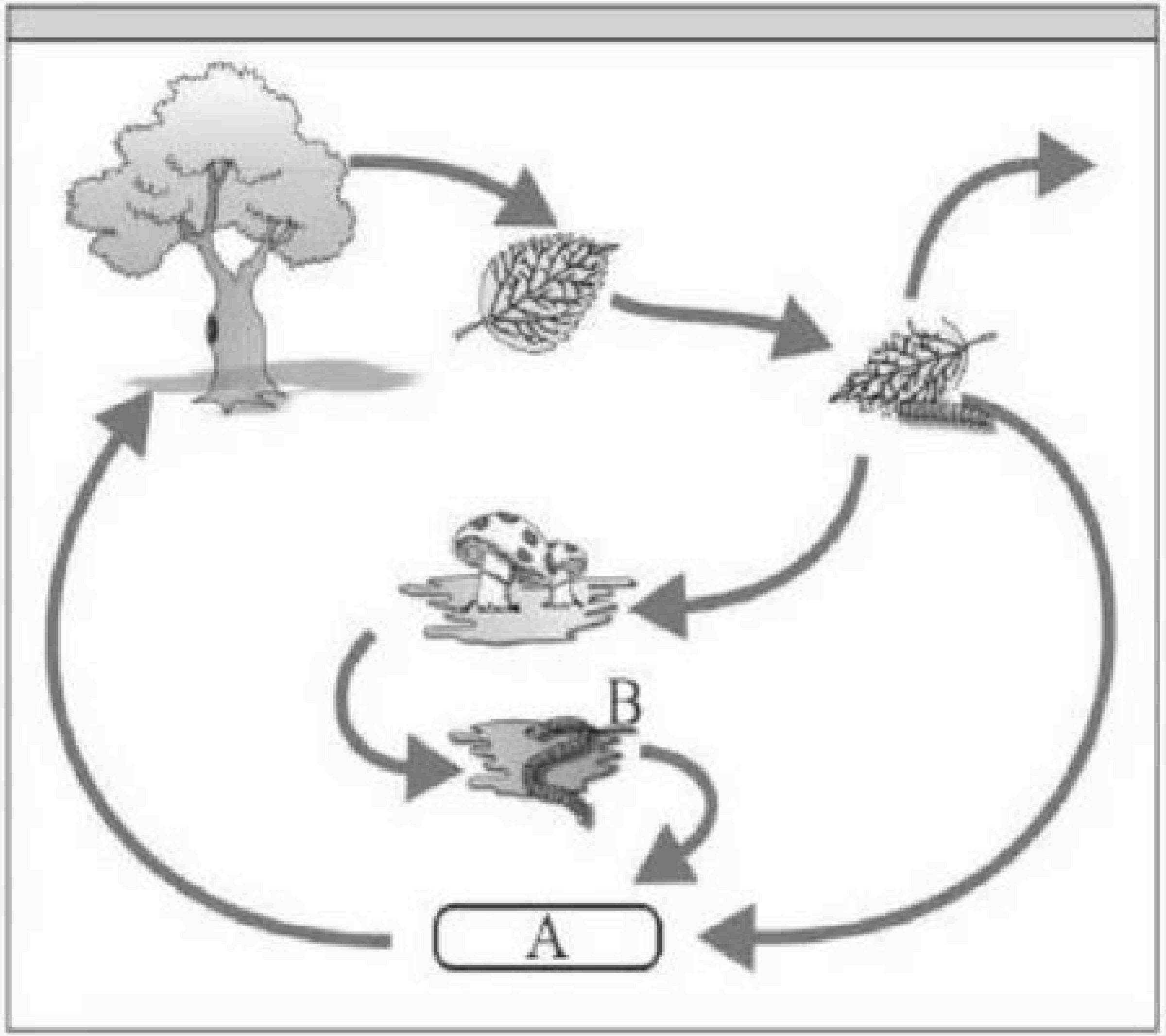
38. Identify A, B

(a) A – Organic rich soil B – Decomposers

(b) A – Inorganic soil B – Producers

(c) A – Organic rich soil B – Producers

(d) A – Organic rich soil B – Consumers



Title: Ecosystem

39. Dead remains of animals including faecal matter, drop over soil and dried plant parts like leaves, bark, flowers etc constitute

i. Litter fall

ii. Below ground detritus

iii. Above ground detritus

(1) i and iii (2) ii and iii

(3) Only ii (4) i and ii

Title: Ecosystem

40. The process of decomposition delays when:

(1) The detritus is made up of sugars and nitrogen compounds

(2) Aeration is sufficient

(3) Warm and moist environment exists

(4) Detritus is rich in lignin and chitin

Title: Ecosystem

41. Leaching, fragmentation and catabolism are some of the important steps of decomposition. Select the correct options w.r.t. the given statements

i. Water soluble inorganic nutrients go into the soil and get precipitated as unavailable salts.

ii. Detritivores breakdown detritus into smaller particles

iii. Decomposers excrete digestive enzymes and degrade detritus into simple inorganic substances.

	Leaching	Fragmentation	Catabolism
(1)	ii	i	iii
(2)	iii	i	ii
(3)	i	ii	iii
(4)	i	iii	ii

42. Choose the correct statements from the following

I. A man eating curd belongs to 3rd trophic level

II. A sparrow feeding on a herbivorous insect is a secondary consumer

III. Rabbit and deer belong to the same trophic level

(1) I, II, III only

(2) I, II only

(3) II, III only

(4) I only

Title: Ecosystem

43. Food chain in which micro organisms break down the food formed by primary producers is

(1) Predator Food Chain

(2) Consumer Food Chain

(3) Detritus Food Chain

(4) Parasitic Food Chain

44. Death of organism is the beginning of the____

(1) Detritus food chain/web

(2) Grazing food chain

(3) Food web

(4) Sea food chain

Title: Ecosystem

45. Snake feed on frog, Hawk feed on Snake. What is the place of Snake in food chain?

(1) Primary Consumer

(2) Secondary Consumer

(3) Tertiary Consumer

(4) Decomposer

46. Ecosystems____

(1) They need a constant supply of energy to synthesise the molecules and to counteract the universal tendency toward increasing disorderliness

(2) They dont require constant supply of energy

(3) Exempt from the Second Law of thermodynamics.

(4) Both 1 and 3

Title: Ecosystem

47. Assertion : The detritus food chain (DFC) begins with dead organic matter.

Reason : Bacteria and fungi meet their energy and nutrient requirements by degrading dead organic matter or detritus.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

Title: Ecosystem

48. Study the following statements about food chain and choose the correct option

I. The length of grazing food chains is general limited to 3-4 trophic levels due to energy loss

II. Removal of most of the carnivores resulted in an increased population of deers

III. The length of food chains may vary from 2-8 trophic levels

IV. Removal of 80% tigers from an area resulted in greatly increased growth of vegetation.

(1) I and II (2) II and III

(3) I and III (4) II and IV

Title: Ecosystem

49. A food web is more realistic representation of feeding relationships than a food chain because

(1) Food web indicates the efficiencies of energy transfer of different trophic levels

(2) Ecological pyramids accommodate food web easily

(3) First law of thermodynamics is depicted by food web but not food chain

(4) One prey species may be eaten by more than one predator species and one predator species may eat more than one prey species

Title: Ecosystem

50. Select the correct statement from the following.

(1) Herbivores are secondary producers in an ecosystem

(2) The mass of living material at a particular time in a trophic level is referred to as standing state

(3) The difference between GPP and NPP is the dfraction of PAR that plants cannot capture

(4) Net primary productivity of oceans is much higher than that of land because 70% of carbon is dissolved in oceans

51. Select the correct answer

A. Detritus food chain may be connected with the grazing food chain at some levels:

some of the organisms of DFC are prey to the GFC animals

B. The natural interconnection of food chains make it a food web.

(1) Only A is correct

(2) Only B is correct

(3) Both are correct

(4) None are correct

52. Which of the following organisms are producers?

(1) Plants and phytoplanktons

(2) Plants and zooplanktons

(3) Zooplanktons and phytoplanktons

(4) Phytoplanktons and detritivores

53. Organisms that depend on plants (directly or indirectly) for food are called

(1) Decomposers

(2) Producers

(3) Consumers

(4) Grazers

Title: Ecosystem

54. Standing crop is -

i. Biomass in a unit area

ii. Number of living organism in unit area

iii. Amount of nutrient such as carbon, nitrogen, calcium in unit area

iv. Amount of Detritus in unit area

(1) Only i
(2) i & ii
(3) i, ii, iii
(4) i, ii, iv

Title: Ecosystem

55. Choose the incorrect option.

(1) The detritus food chain (DFC) begins with dead organic matter.

(2) In an aquatic ecosystem, GFC is the major conduit for energy flow.

(3) GFC (Grazing Food Chain) begins with producers at the first trophic level

(4) In an aquatic ecosystem, DFC is the major conduit for energy flow.

56. Statement A : A detritivore is animal feeding on dead and decaying organic matter.

Statement B : In a pond, phytoplankton synthesise food materials from dissolved nutrients by photosynthesis.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

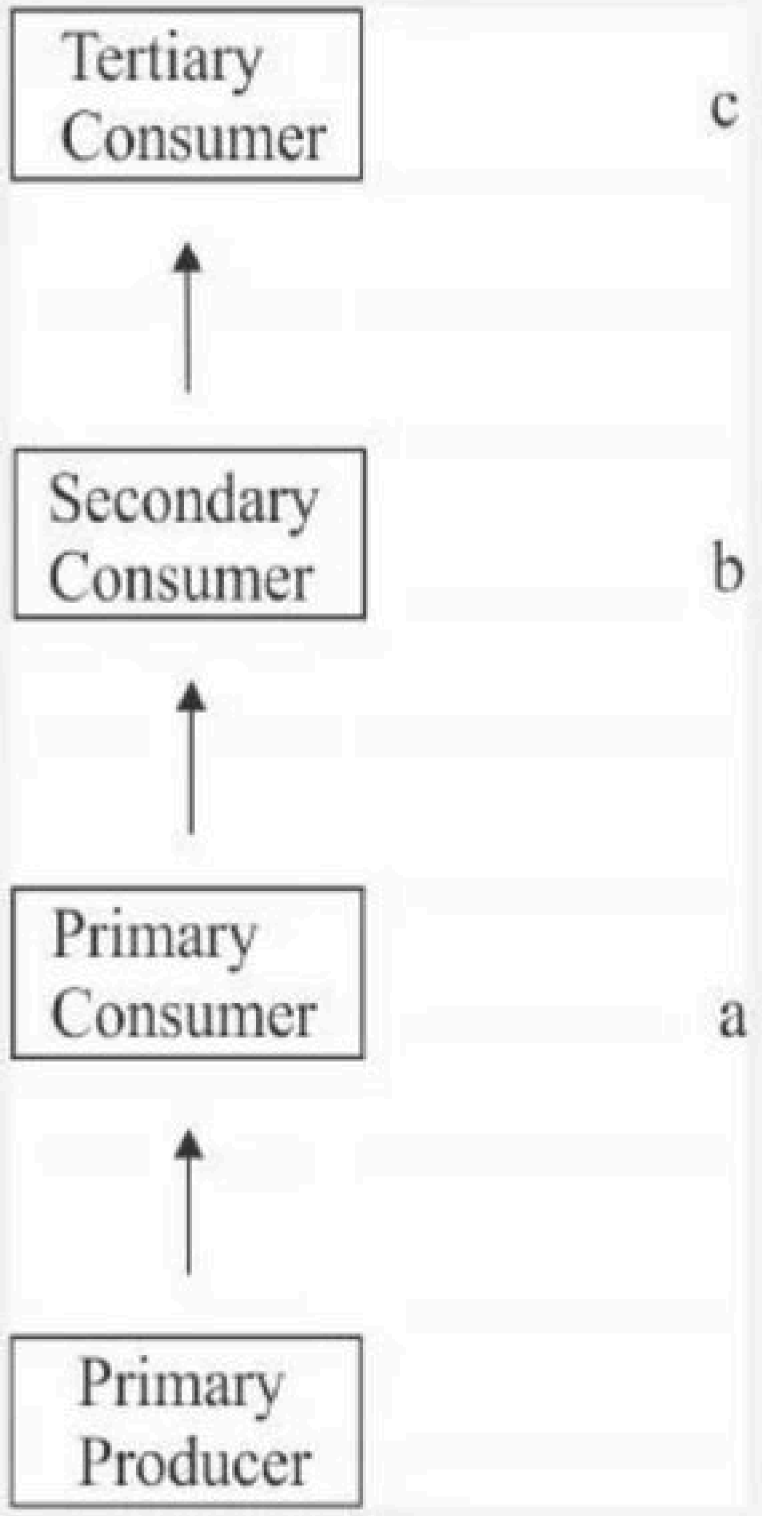
57. Identify a, b, c

(1) a-Carnivore, b-herbivore, c-Top carnivore

(2) a-Top carnivore, b-herbivore, c-Carnivore

(3) a-herbivore, b-Carnivore, c-top carnivore

(4) a-top carnivore, b-Carnivore, c-herbivore



Title: Ecosystem

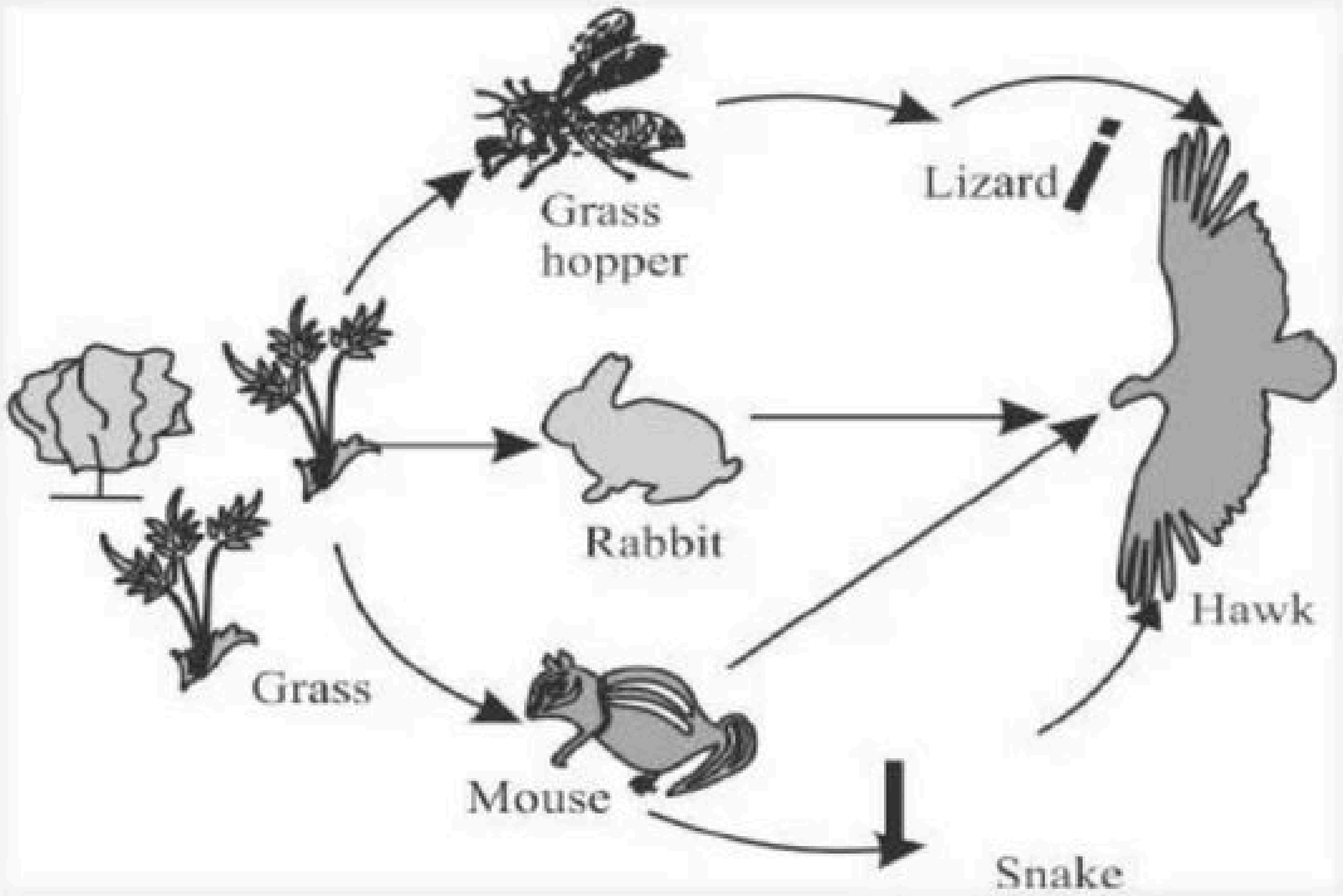
58. In the food web given, increase in population of hawks will not result in

(1) Decrease in the population of grasshoppers

(2) Increase in the population of producers

(3) Decrease in the population of lizards

(4) Decrease in the population of rabbits



Title: Ecosystem

59. Study the following statements

I. Removal of carnivores resulted in increased herbivores

II. Removal of phytoplanktons resulted in decreased productivity in a lake

III. Removal of tigers from an area resulted in increased growth of vegetation

IV. Removal of decomposers resulted in decreased primary productivity

(1) Only one statement is correct

(2) Only two statements are correct

(3) Only three statements are correct

(4) All are correct

Title: Ecosystem

60. If 20 J of energy is trapped at producer level, then how much energy will be available to peacock as food in the following chain?

Plant → Mice → Snake → Peacock

(1) 0.02 J

(2) 0.002 J

(3) 0.2 J

(4) 0.0002 J

Title: Ecosystem

61. Statement A : Death of organism is the end of the detritus food chain/web.

Statement B : No energy that is trapped into an organism remains in it for ever.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

62. Assertion : Plants capture only 2-10 per cent of the PAR and this small amount of energy sustains the entire living world.

Reason : Less than 50 per cent of incident solar energy is photosynthetically active radiation (PAR).

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

Title: Ecosystem

63. The first trophic level in grazing food chain is always a green plant because

(1) They along have capability to fix atmospheric N_2

(2) Rich in chlorophyll

(3) Of wide distribution

(4) Their capacity to convert light energy to chemical energy

Title: Ecosystem

64. Choose the correct sequence

(1) Fish → crustaceans → zooplankton → phytoplankton

(2) Phytoplankton → zooplankton → crustaceansfish

(3) Crustaceans → fishzooplankton → phytoplankton

(4) Zooplankton → phytoplankton → crustaceans → fish

Title: Ecosystem

65. Match the trophic levels with their correct species in grassland ecosystem.

Select the correct option :

Column I	Column II
A Fourth trophic level	P Crow
B Second trophic level	Q Vulture
C First trophic level	R Rabbit
D Third trophic level	S Grass

(1) A - R, B - Q, C - P, D – S

(2) A - S, B - R, C - Q, D - P

(3) A - P, B - Q, C - R, D – S

(4) A - Q, B - R, C - S, D - P

66. Which of the following statements about food chains and energy flow through ecosystem is false?

(1) A single organism can feed at several trophic levels

(2) Detritivores feed at all trophic levels except the producer level.

(3) The lower the trophic level at which an organism feeds, the more energy available

(4) Those who are not producers are called heterotrophs.

Title: Ecosystem

67. Statement A : Energy at a higher trophic level is always more than at a lower level.

Statement B : The pyramid of biomass in sea is generally inverted.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

Title: Ecosystem

68. Which of the following statements is not correct?

(1) Pyramid of biomass in sea is generally inverted.

(2) Pyramid of biomass in sea is generally upright

(3) Pyramid of energy is always upright

(4) Pyramid of numbers in a grassland ecosystem is upright.

Title: Ecosystem

69. Mr.X is eating curd/yoghurt. For this food intake in a food chain he should be considered as occupying

(1) Second trophic level

(2) First trophic level

(3) Fourth trophic level

(4) Third trophic level

Title: Ecosystem

70. Pyramid of energy is always upright which indicates that

(1) Producers have lowest energy conversion efficiency

(2) Energy conversion efficiency is same for all trophic levels.

(3) Herbivores have better energy conversion efficiency than carnivores.

(4) Carnivores have better efficiency than herbivores

71. To identify how many organisms are present at each trophic level, the pyramid is called

(1) An energy flow pyramid

(2) Pyramid of numbers

(3) Pyramid of energy

(4) Food chain/food web pyramid

Title: Ecosystem

72. Read the following statements and select the correct option

I. Producers constitute the first trophic level of a detritus food chain

II. Productivity of an aquatic system is less than that of a terrestrial system

III. A given species may occupy more than one trophic level in the same ecosystem at the same time

(1) II and III (2) I and II

(3) Only III (4) I and III

Title: Ecosystem

73. Which of the following statement is correct?

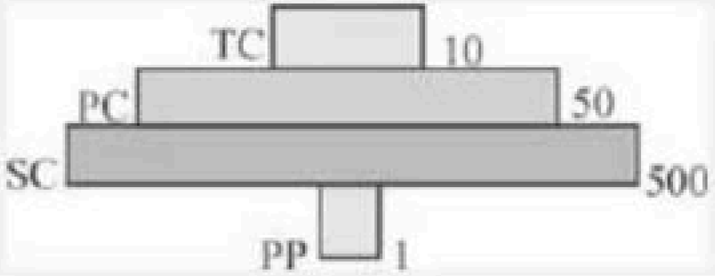
(1) Pyramid of number in grassland ecosystem is upright.

(2) Pyramid of number in aquatic ecosystem is inverted.

(3) Pyramid of biomass in tree ecosystem is inverted.

(4) Pyramid of biomass in aquatic ecosystem is upright

74. Given below is an imaginary pyramid of numbers. What could be one of the possibilities about certain organisms at some of the different levels?



(1) Level PC is "insects" and level SC is "small insectivores birds".

(2) Level PP is "phytoplanktons" in sea and "whale" on top level TC

(3) Level PP is "pipal trees" and the level is "sheep"

(4) Level PC is "rats" and level SC is "cats".

Title: Ecosystem

75. The pyramid of biomass in marine ecosystem is generally inverted because

(1) Primary productivity of marine ecosystems is lesser than that of terrestrial ecosystems

(2) The size of individual organisms decreases from lower trophic levels to higher trophic levels

(3) Compared to primary consumers, phytoplankton synthesize less biomass per unit time per unit area

(4) The standing crop of phytoplankton at any given moment is lower than that of primary consumers

76. Statement A : Pyramid of energy may be upright or inverted.

Statement B : Only 20% of energy goes to next trophic level.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

77. Assertion : The pyramid of biomass in sea is generally inverted.

Reason : Number of phytoplanktons is more than number of fishes.

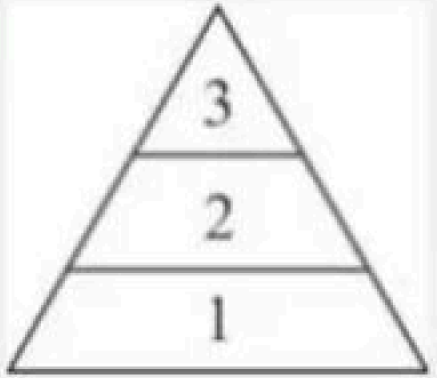
(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

78. In the given pyramid of energy, which of the following can be placed in first trophic level?



(1) Herbivores
(2) Consumer
(3) Plants
(4) All the above

Title: Ecosystem

79. What type of pyramid would be obtained with the following data?

A. Secondary consumers: 2.6 g/m^2/ year

B. Primary consumers: 48 g/m^2/ year

C. Primary producers: 763 g/m^2/ year

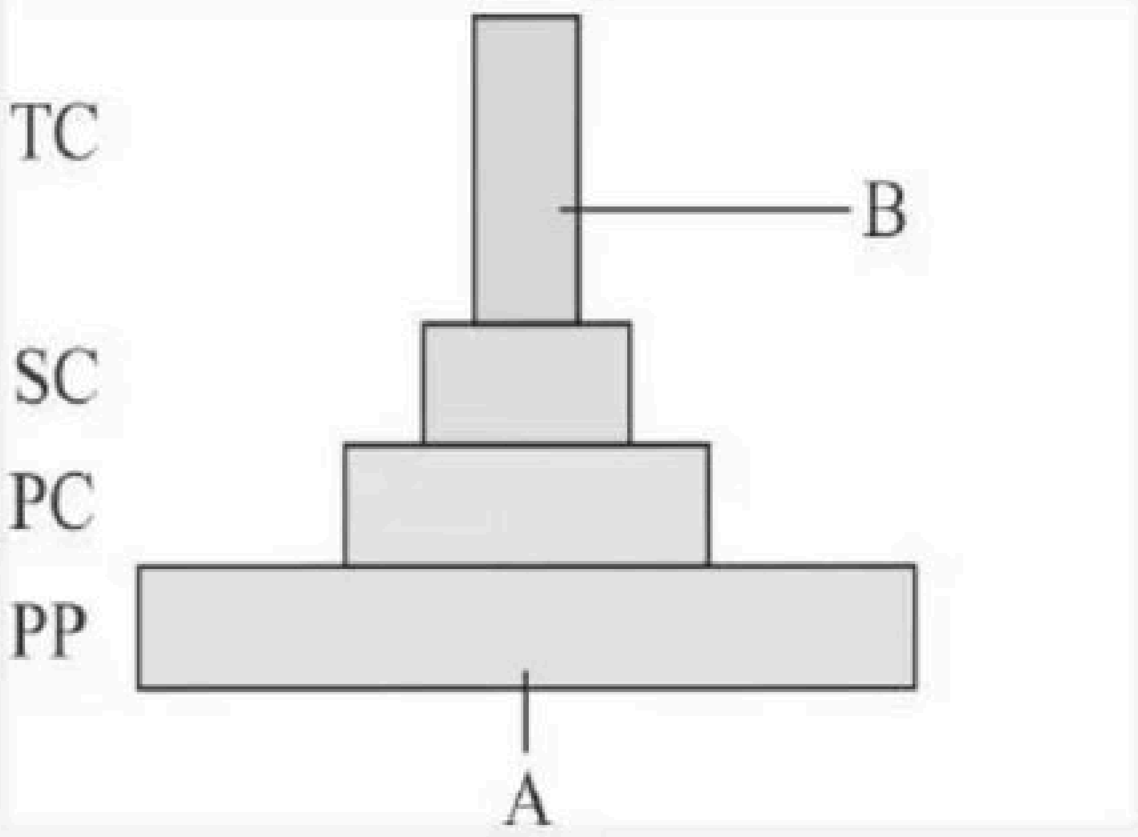
(1) Upright pyramid of energy

(2) Inverted pyramid of energy

(3) Upright pyramid of biomass

(4) Inverted pyramid of biomass

80. Find out the labelled parts



(1) A – 1, 000, 000 J B – 10 J

(2) A – 1, 000, 000JB – 100J

(3) A – 1, 000, 00 J B – 10 J

(4) A – 1, 000, 00 J B – 100 J

81. Statement A : The three types of ecological pyramids that are usually studied are (a) pyramid of number; (b) pyramid of biomass and (c) pyramid of energy.

Statement B : Organism may occupy more than one trophic level simultaneously.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

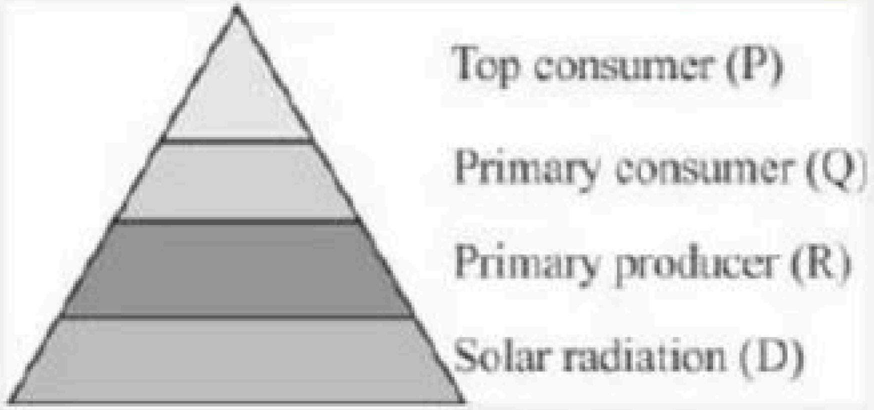
(4) Both Statements are incorrect.

Title: Ecosystem

82. (P) 10kcal/m^2/yr **(Q) 100kcal/m^2/yr**

(R) 1000kcal/m^2/yr **(S) 1, 00, 000kcal/m^2/yr**

In the above energy pyramid, rate of primary production is



(1) 1000 kcal/m^2/yr (2) 110 kcal/m^2/yr

(3) Uncertain (4) 100 kcal/m^2/yr

Title: Ecosystem

83. Which of the following is not used for construction of an ecological pyramid?

(1) Dry weight

(2) Rate of energy flow

(3) Individuals

(4) Fresh weight

84. Match the correct option

Column I	Column II
A Saprotrophs	P Primary Consumer
B Lion	Q Producer
C Plants	R Decomposer
D Herbivores	S Secondary consumer

(1) A – S, B – P, C – Q, D – R
(2) A – S, B – P, C – Q, D – R
(3) A – R, B – S, C – Q, D – P
(4) A – S, B – P, C – R, D – 2Q

85. Match the columns I and II.

Column I		Column II	
A	Detritus break down	P	Catabolism
B	Inorganic nutrients go down into the soil horizon	Q	Mineralisation
C	Detritus is broken by bacteria and fungi	R	Leaching
D	Formation of colloidal matter	S	Fragmentation
E	Release of inorganic nutrients	T	Humification

(1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S

Title: Ecosystem

86. Match the columns I and II wrt the pond.

Column I		Column II	
A	Consumers	P	Bacteria and flagellates
B	Decomposers	Q	Zooplankton
C	Autotrophs	R	Phytoplankton

(1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R

87. Match the columns I and II.

Column I		Column II	
A	Standing crop	P	Amount of living matter present in an ecosystem
B	Standing state	Q	Nitrogen
C	Gaseous cycles	R	Amount of nutrients present in an ecosystem
D	Sedimentary cycle	S	Phosphorous

(1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S

88. In a pyramid, carnivores cannot occupy

(1) 3 trophic level

(2) 3 and 4 trophic levels

(3) 1,2 trophic levels

(4) 4 and 5 trophic levels

Title: Ecosystem

89. Vertical distribution of different species occupying different levels in a biotic community is known as

(1) Stratification

(2) Zonation

(3) Pyramid

(4) Divergence

90. Pyramid of numbers is:

(1) Always upright

(2) Always inverted

(3) Either upright or inverted

(4) Neither upright nor inverted.

91. The green plants in an ecosystem which can trap solar energy to convert it into chemical bond energy are called

(1) Producer

(2) Autotrophs

(3) Consumer

(4) Both 1 and 2

92. Example of man-made ecosystem are?

(1) Crop field

(2) Aquarium

(3) Desert

(4) Both 1 and 2

Title: Ecosystem

93. The process of mineralisation by micro organisms helps in the release of:

(1) Inorganic nutrients from humus

(2) Both organic and inorganic nutrients from detritus

(3) Organic nutrients from humus

(4) Inorganic nutrients from detritus and formation of humus.

94. Transfer of energy follows a percent law . What does it state?

(1) Only 10 per cent of the energy is transferred to each trophic level from the higher trophic level.

(2) Only of 50% the energy is transferred to each trophic level

(3) Only of 20% the energy is transferred to each trophic level

(4) Only 10 per cent of the energy is transferred to each trophic level from the lower trophic level.

95. What is the trophic level of rat in a food chain?

(1) T1

(2) T2

(3) T3

(4) T4

Title: Ecosystem

96. The organic substances, which decompose slowly are

(1) Chitin and Lignin

(2) Rich in nitrogen

(3) Rich in water-soluble substances like sugars

(4) All of these

Title: Ecosystem

97. Primary production is

(1) Expressed in terms of weight (gm^{-2}) or energy ($kcal\ m^{-2}$)

(2) The amount of biomas or organic matter produced per unit area over a time period by plants during photosynthesis

(3) The amount of inorganic matter produced per unit area over a time period by plants during photosynthesis

(4) Both 1 and 2

98. Which one of the following is not characteristic of humus?

(1) It is further degraded by the process of humification

(2) It is colloidal in nature and serves as a reservoir of nutrients

(3) It is rich in organic matter such as cellulose and lignin'

(4) More than 1 option is correct

99. Statement A : Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.

Statement B : A constant inputof solar energy is the basic requirement for any ecosystem to function and sustain.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.

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100. Decomposers are very important in the ecosystems because they

(1) Transduce light energy into chemical energy

(2) Return energy from higher trophic levels back to the producers

(3) Convert organic matter in the detritus into inorganic compounds usable by primary producers

(4) Synthesize organic compounds from inorganic compounds

THANK YOU

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